FIT2179 Data Visualisation

Course Notes

PROPORTIONAL SYMBOL MAPS

13 September 2026

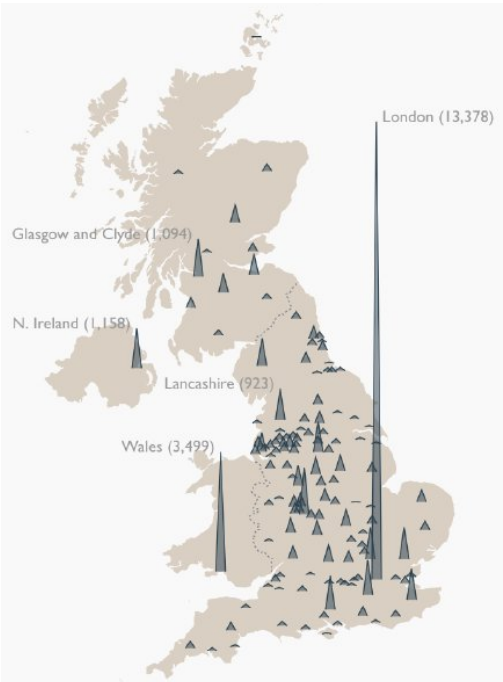
PROPORTIONAL SYMBOL MAPS

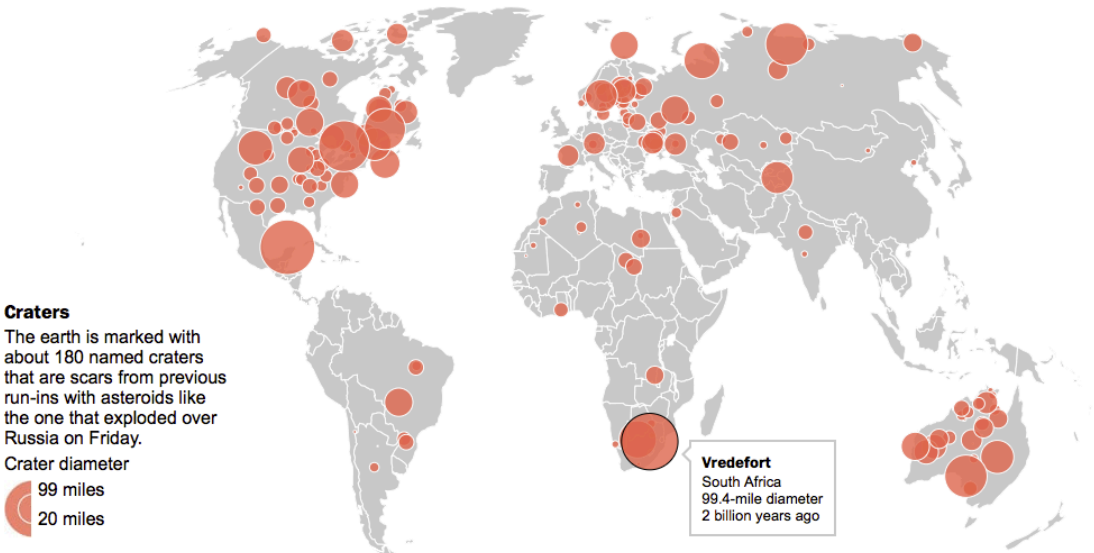
Proportional symbol maps are a good choice when points or areas carry a quantitative attribute.

The map on the right uses length-proportional bars to show Covid-19 cases, where the attribute belongs to an area geometry.

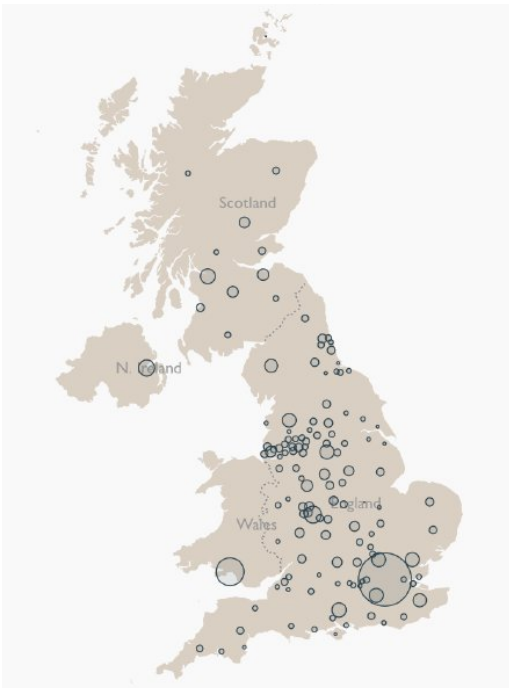
The map below uses area-proportional circles to show the size of craters, where the attribute belongs to a point geometry. Size is the channel used in both cases. In the first map the marks are triangles, though lines are more common; in the second they are circles, though other shapes such as area-proportional squares are equally possible.

Proportional symbol maps can show a few dozen items. Beyond that, the symbols heavily overplot and the map becomes difficult to read.





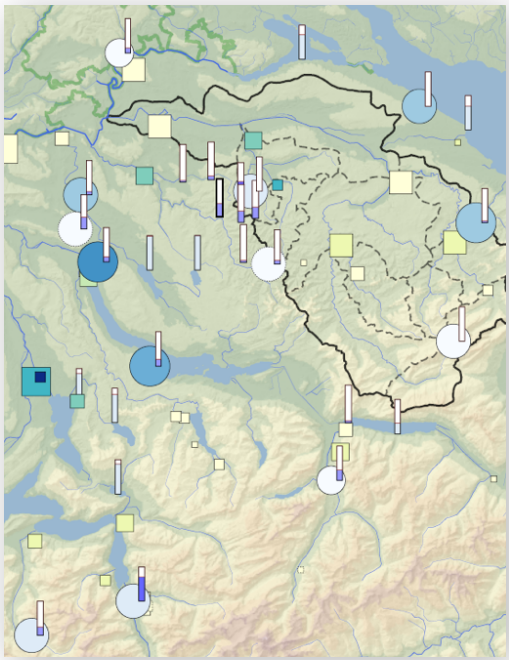
Map by The Washington Post,



Compare this map with the one on the previous page. The length-proportional bars are easier to compare, and the spike for London is more striking than the area-proportional circles. However, the bars would not be possible if London sat at the top edge of the map, because the spike would no longer fit.

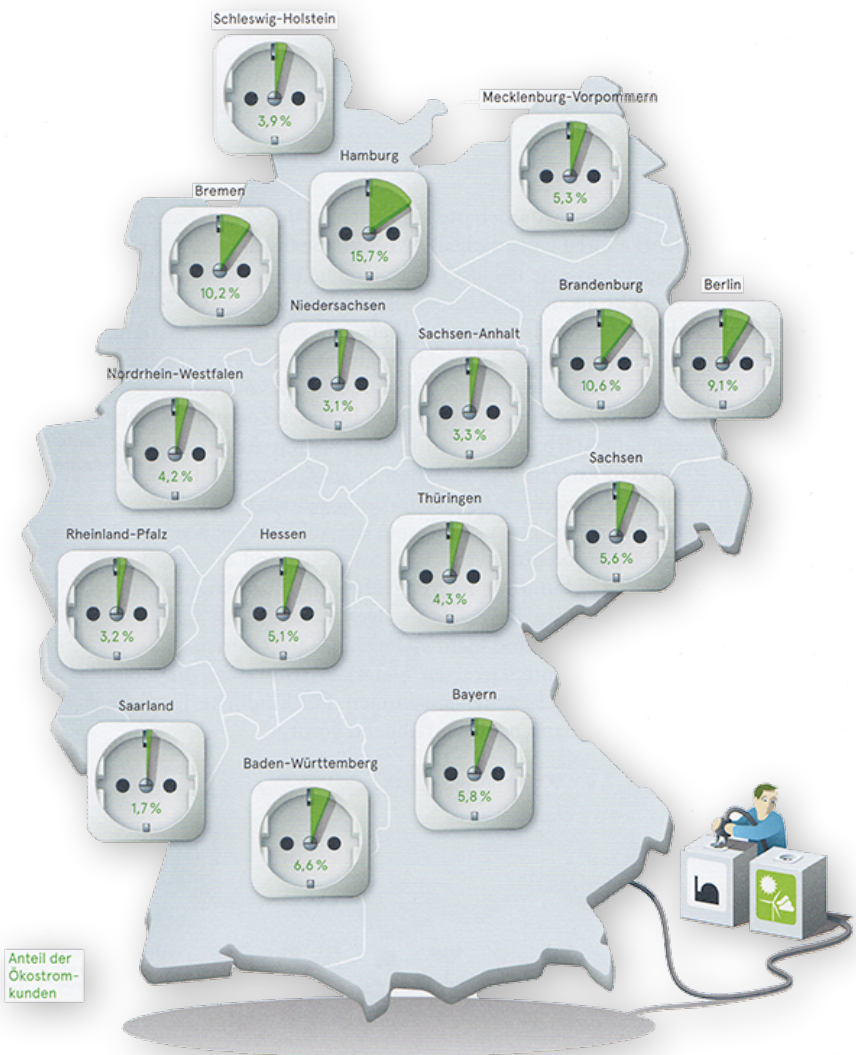
Map from Slocum et al. 2022, Thematic Cartography and Geovisualization, ed. 4.

The hydrology example shows water levels in rivers and lakes. Its bars are framed, which is more effective than unframed bars. Circles and squares serve as area-proportional symbols, and colour encodes additional attributes.



Map by C. Lienert, ETH Zurich

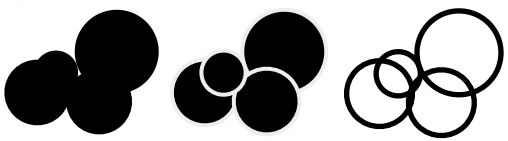
Other symbol types are possible, such as stacked bar charts or pie charts, as on the map showing the percentage of alternative energy customers in Germany a few years ago.



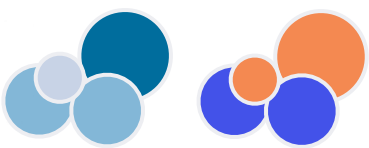
Map from
Stolz, M. & Block, J. Deutschlandkarte, 2012, Zeit Magazin

DESIGN PRINCIPLES

Overlapping symbols are possible. The readability of overlapping symbols can be increased by applying strokes to symbols (centre) or making symbols (partially) transparent (right).



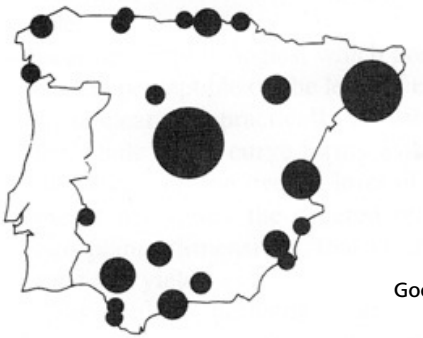
Colour can be used for double encoding of the same attribute, or to show a second attribute (as shown by the hydrology map on the previous page).



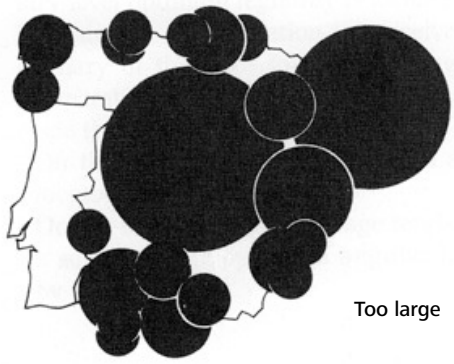
The size of symbols requires consideration. Symbols that are too small make values hard to compare; symbols that are too large overplot, hiding the geographic position of the symbols and making comparison of values difficult.



Too small



Good



Too large